

Technical drawing of a 2-phase hybrid stepper motor (SMH2405) showing side and front views with dimensions and specifications.

Dimensions (mm):

- Side View: Total length 100±1, mounting flange width 21±1, flange thickness 1.6, mounting hole diameter 19±0.2, shaft diameter 38.1⁰_{-0.05}, mounting hole offset Y, mounting hole diameter 7, mounting hole offset 50.
- Front View: Mounting flange outer diameter 60max, mounting hole diameter 47.14±0.2, mounting hole offset 47.14±0.2, mounting hole diameter 60max, mounting hole offset 1500min, mounting hole diameter 4-φ5.2, mounting hole offset 1500min, mounting hole diameter 1500min.

Specifications:

Item	Value
相数 (Number of Phases)	2
基本步距角 (Basic Step Angle)	1.8°
基本步距角误差 (空载) (Basic Step Angle Error (No Load))	±0.09°
额定电流 (Rated Current)	5.0 A
电阻 (相) (Resistance (Phase))	0.67 Ω±10%
电感 (相) (Inductance (Phase))	2.3 mH±20%
保持力矩 (Holding Torque)	3.7 N.m
径向跳动 (Radial Runout)	0.025 MM (负载450g)
轴向跳动 (Axial Runout)	0.075 MM (负载920g)
额定工作状态 (Rated Working State)	连续 (Continuous)
寿命 (小时) (Life (Hours))	5000H以上
温升 (Temperature Rise)	80 K以下
绝缘耐压 (Insulation Withstanding Voltage)	500V A.C
绝缘电阻 (Insulation Resistance)	100MΩ以上
绝缘等级 (Insulation Class)	B
使用温度范围 (Operating Temperature Range)	-20°C~+50°C
使用湿度范围 (Operating Humidity Range)	20%RH~90%RH
保存温度范围 (Storage Temperature Range)	-20°C~+70°C
保存湿度范围 (Storage Humidity Range)	15%RH~95%RH
重量 (Weight)	1.7 KG

Motor Label Information:

- MODEL: 2405HS0D10
- 2-PHASE 1.8°/STEP
- 品牌 (Brand): HDB
- 额定电流 (Rated Current): 5.0 A
- 电阻 (Resistance): 0.67 Ω
- 电感 (Inductance): 2.3 mH
- 保持力矩 (Holding Torque): 3.7 N.m
- 电机型号 (Motor Model): SMH2405

Wiring Diagram:

- Black: A
- Green: Ā
- Red: B
- Blue: B̄

Revisions:

Revision	Description	Date
1	Initial Release	2022.11.10

Customer Confirmation:

Mark	Quantity	Change File No.	Signature	Date
Design		Standardization		
Check		Process		
Approval		Approval		

Viewing Method:

- SMH2405
- 二相混合步进电机 (2-Phase Hybrid Stepper Motor)
- 比例 (Scale): 1:1
- 深圳市德智高新有限公司 (Shenzhen Deyi High-Tech Co., Ltd.)
- WWW.HDBMOTOR.COM